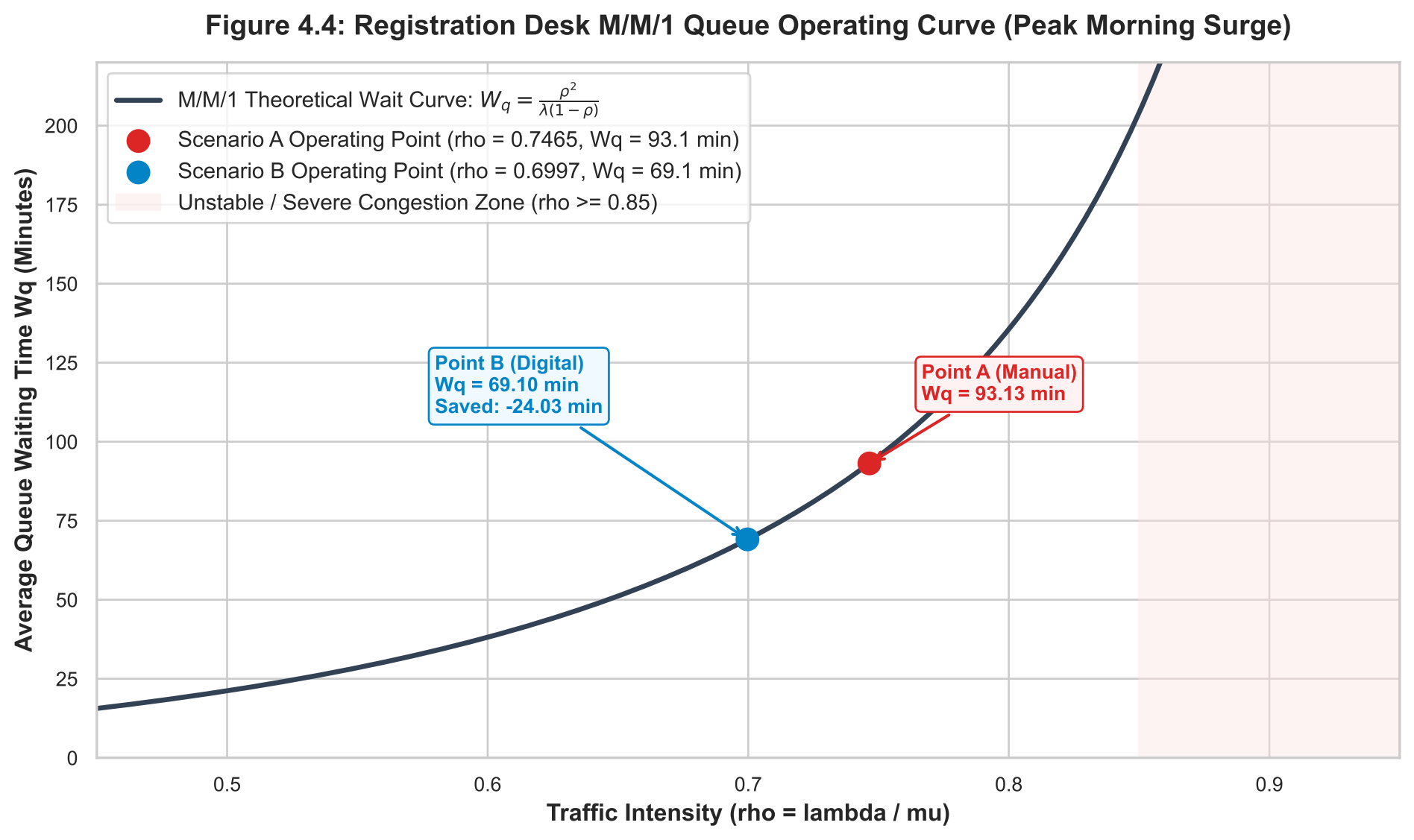
Figure 4.4: Registration Desk M/M/1 Queue Operating Curve (Peak Morning Surge)
M/M/1 Theoretical Wait Curve: $W_q = \frac{\rho^2}{\lambda(1-\rho)}$
Scenario A Operating Point (rho = 0.7465, Wq = 93.1 min)
Scenario B Operating Point (rho = 0.6997, Wq = 69.1 min)
Unstable / Severe Congestion Zone (rho >= 0.85)
Point B (Digital)
Wq = 69.10 min
Saved: -24.03 min
Point A (Manual)
Wq = 93.13 min
Average Queue Waiting Time Wq (Minutes)
0
25
50
75
100
125
150
175
200
0.5
0.6
0.7
0.8
0.9
Traffic Intensity (rho = lambda / mu)